

EVOLUTION OF HELIUM STARS

(Trial models' update)

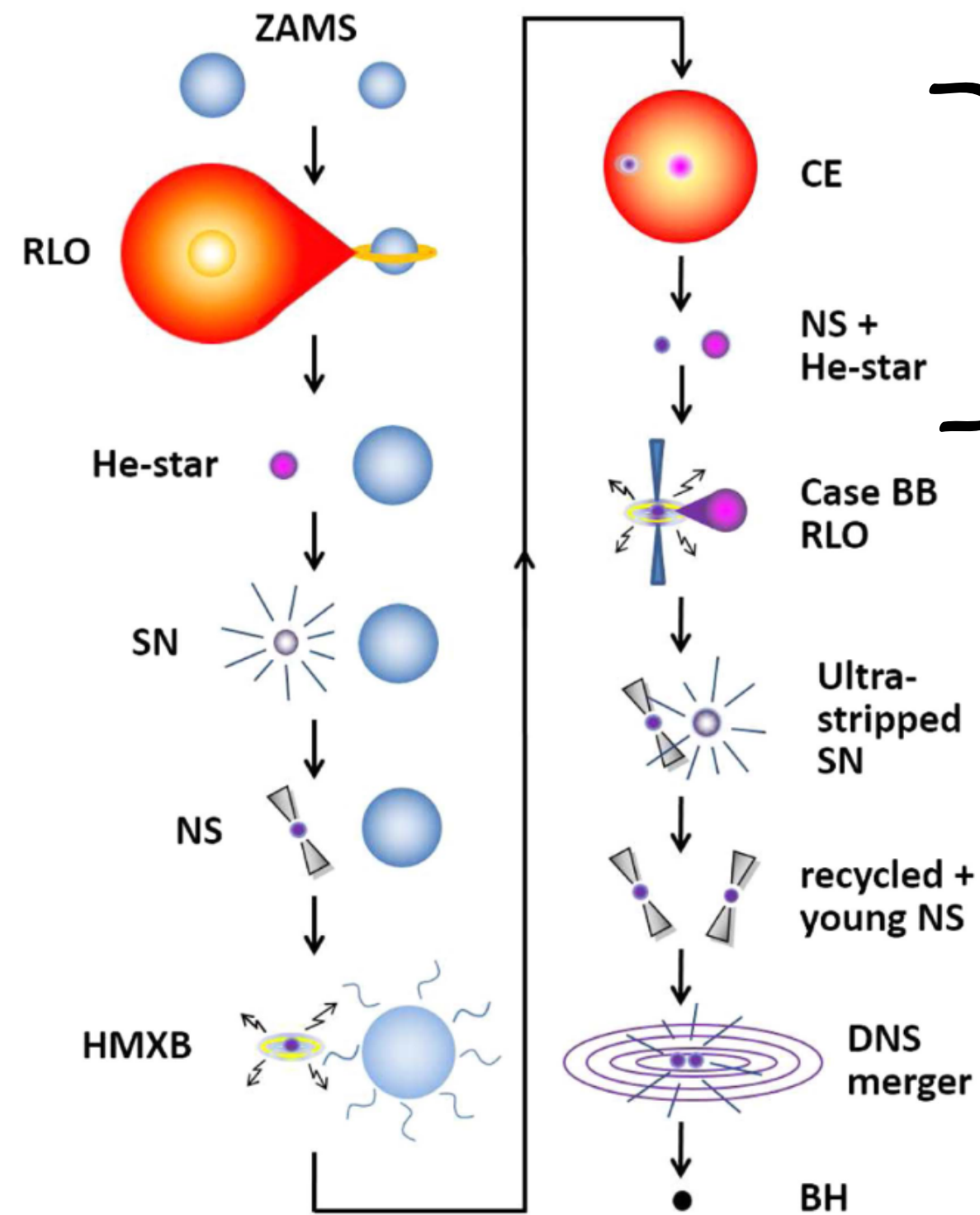


Presented by Ashwathi

Pulsar Coffee 25th Oct'23

Overview

- A little about Helium stars and its role in compact merger formation
- A quick introduction to Modules for Experiments in Stellar Astrophysics (MESA)
- Models evolved using MESA (Results)
 - Low-mass helium star without mass loss
 - Low-mass + high-mass helium star with mass loss
- Errors encountered



Termination of the CE
phase - ejection of
hydrogen envelope

Left behind with a
naked helium star and
NS companion

Figure 1. Illustration of the formation of a DNS system that merges within a Hubble time and produces a single BH, following a powerful burst of GWs and a short GRB. Acronyms used in this figure—ZAMS: zero-age main sequence; RLO: Roche-lobe overflow (mass transfer); He-star: helium star; SN: supernova; NS: neutron star; HMXB: high-mass X-ray binary; CE: common envelope; BH: black hole.

Why is binary evolution studied using helium stars?

- Doing so reduces the computational expense trivially
- Isolates the evolution of the central core from uncertainties in red giant mass loss, semiconvection and overshoot mixing during hydrogen core and shell burning

MESA

- **Modules for Experiments in Stellar Astrophysics (MESA)** - open source, robust, efficient, thread-safe libraries for a wide range of applications in computational stellar astrophysics.
- A 1-D stellar evolution module, MESA star, combines many of the numerical and physics modules for simulations of a wide range of stellar evolution scenarios
- MESA star solves the fully coupled structure and composition equations simultaneously.

Different MESA modules

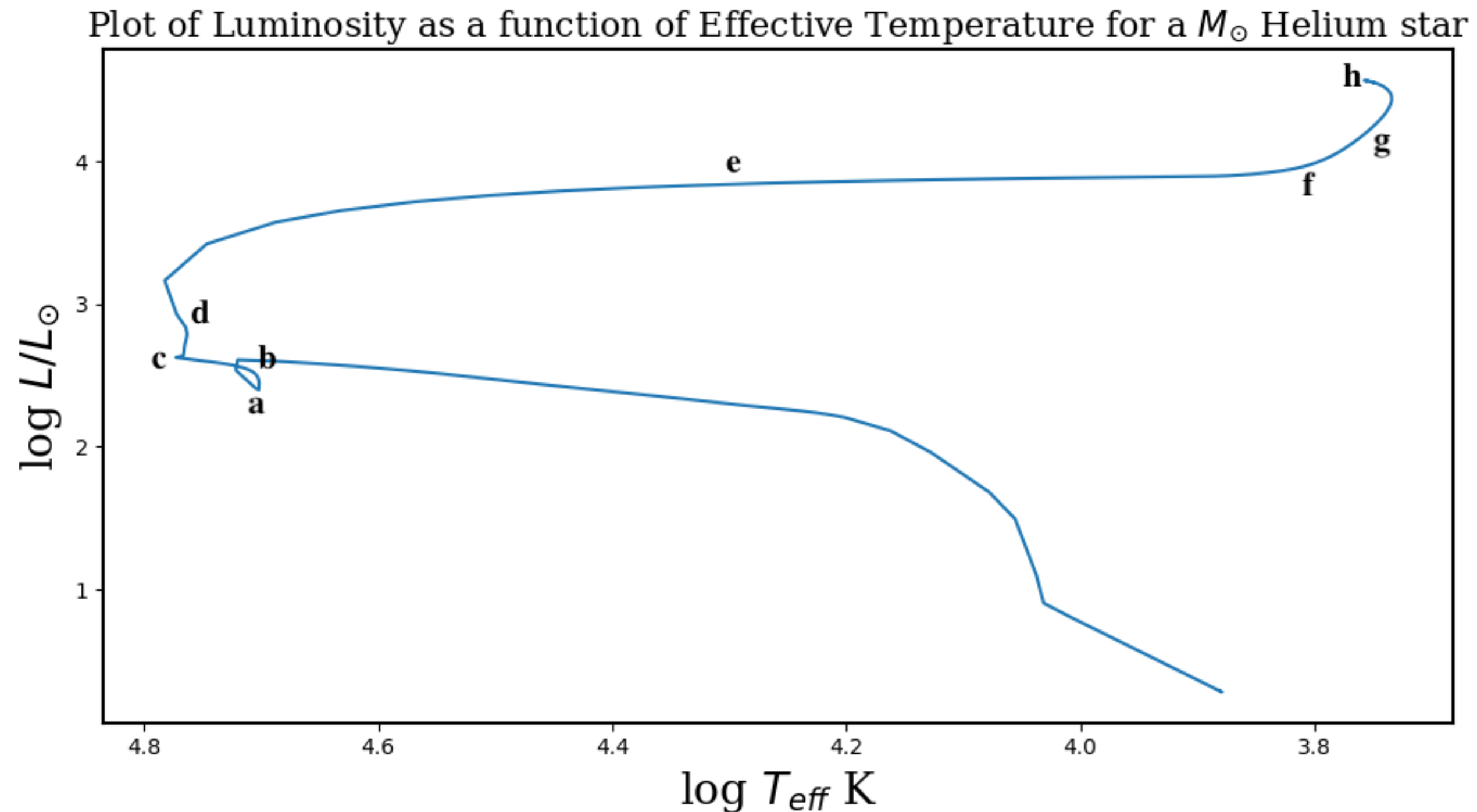
For numerical method

- mtx
- num

For physics

- kap
- EOS
- const
- chem
- rates
- net
- mlt

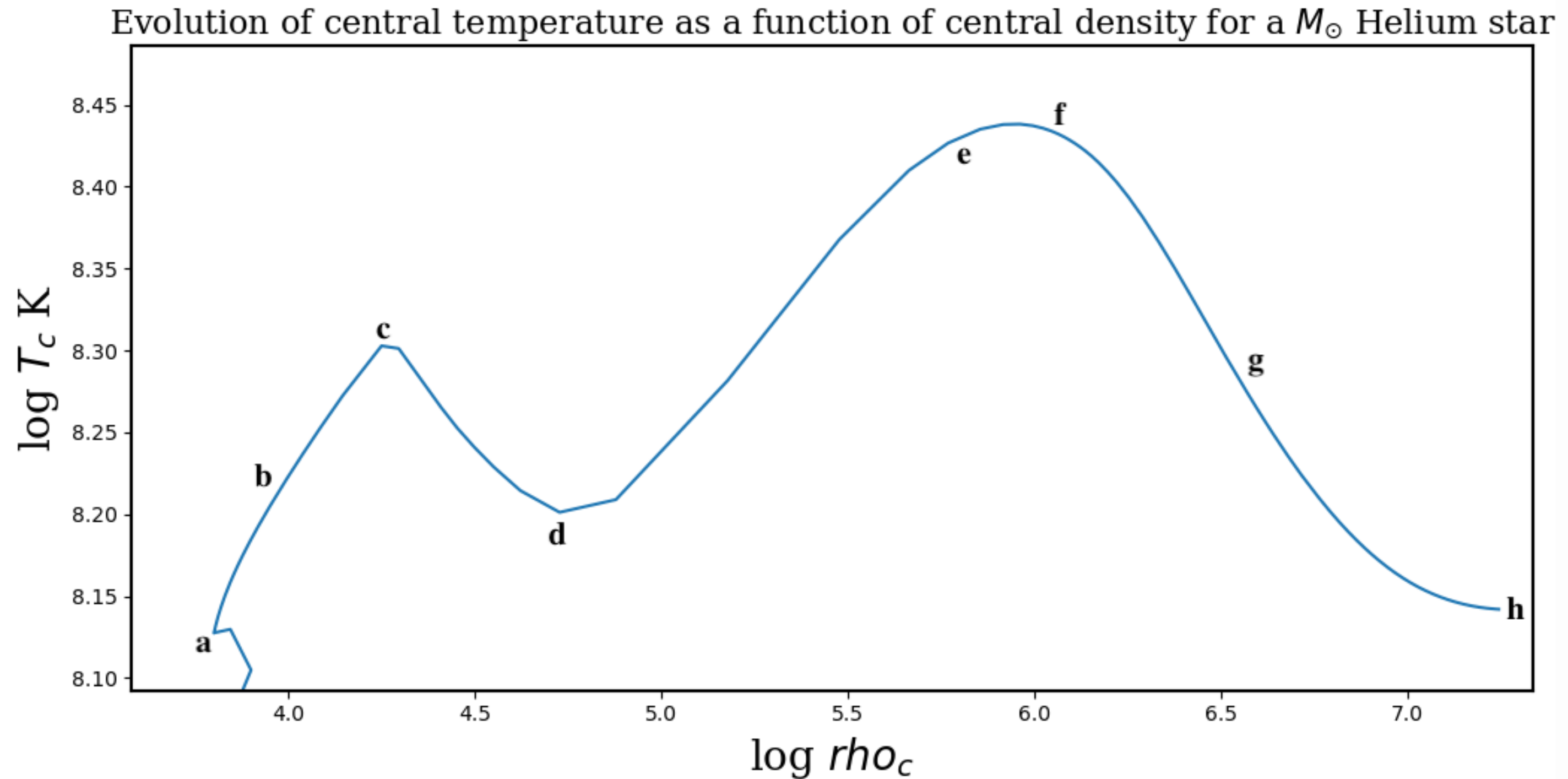
Hertzsprung-Russell Diagram for a solar mass helium star(without mass loss)



The letters marked along the curve give the correspondence of evolutionary stages:

main sequence phase (a-b), exhaustion of central helium (b-c), first cooling phase(c-d), slow phase of shell helium burning(d-e), and fast phase of helium shell burning(e-h)

The evolutionary tracks in the central density–central temperature plane of a solar mass helium star

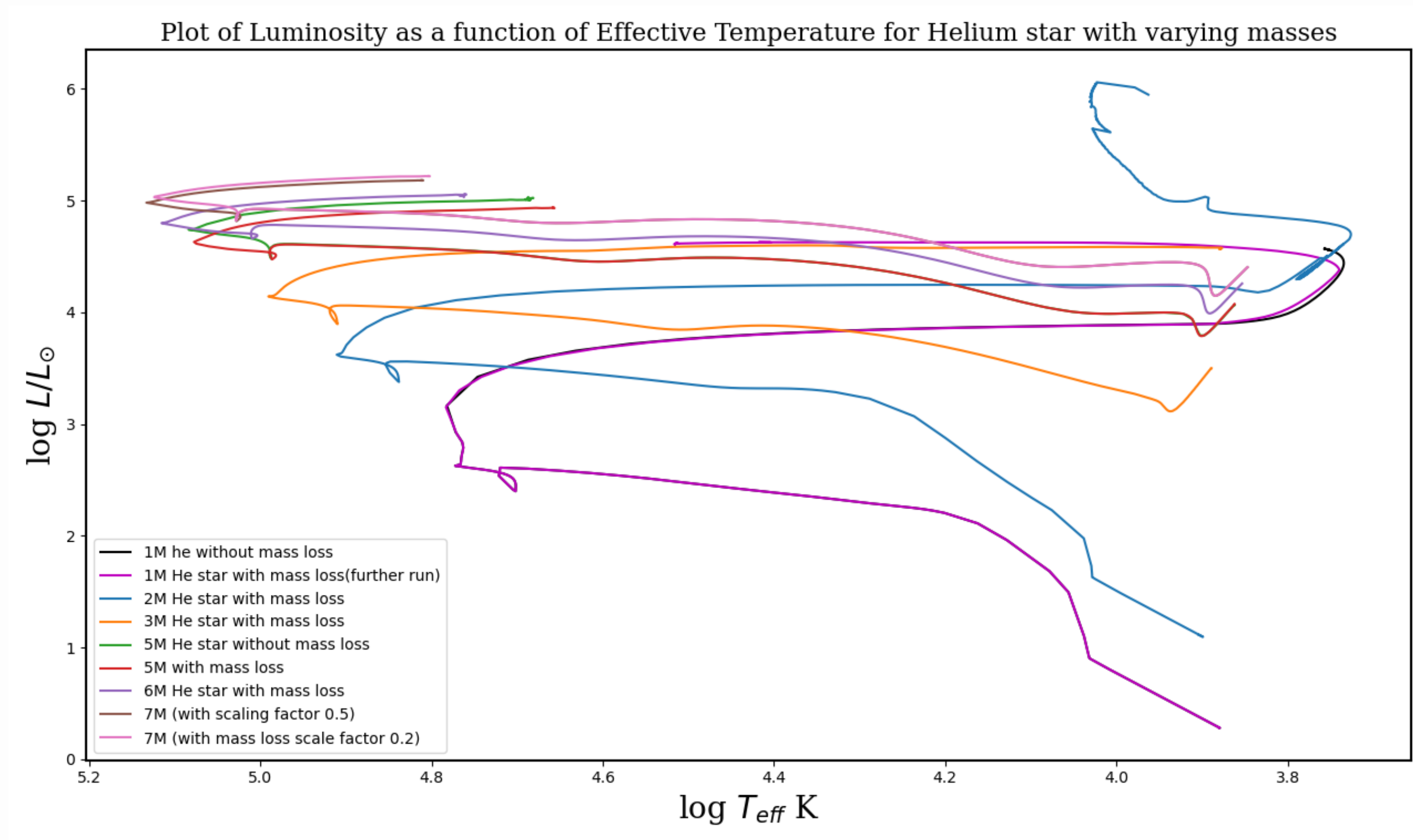


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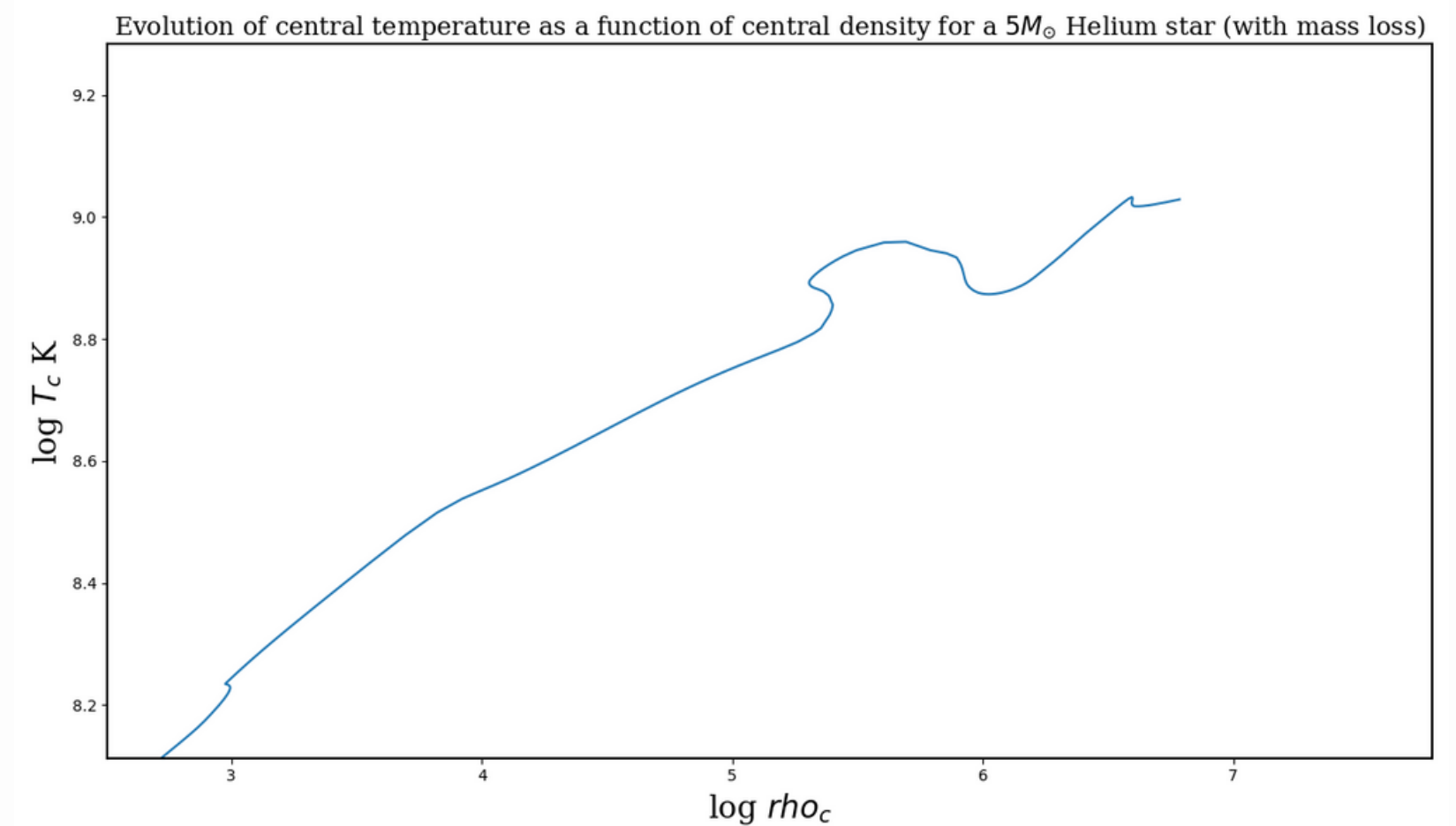
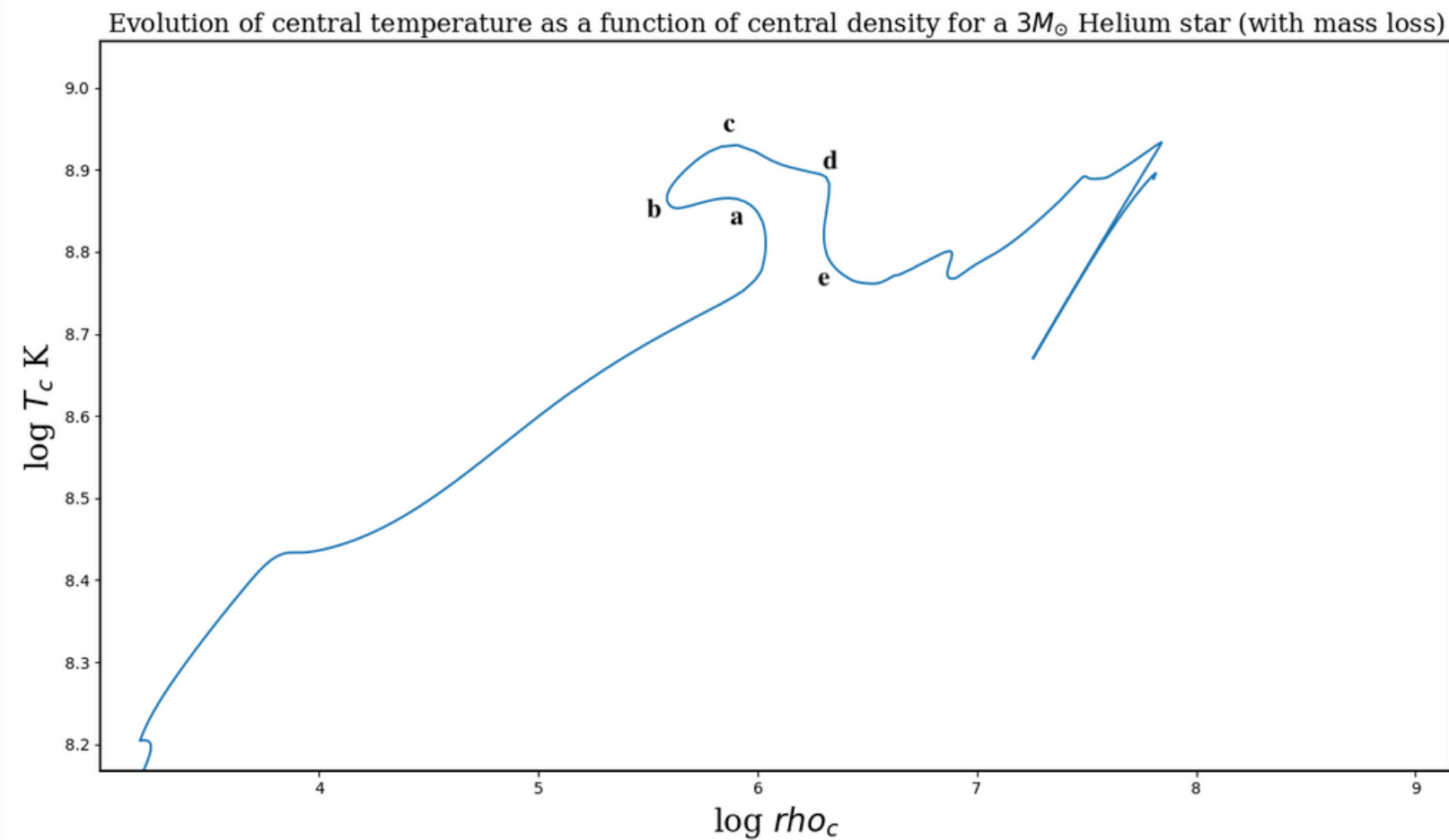
Important piece of Physics included

- Mass loss (by winds)
 - Dutch wind scheme for massive stars
- Mixing length Theory
- Semiconvection
 - Ledoux criteria to determine the position of convective boundaries

Hertzsprung-Russell Diagram for a grid of varying masses of Helium star.

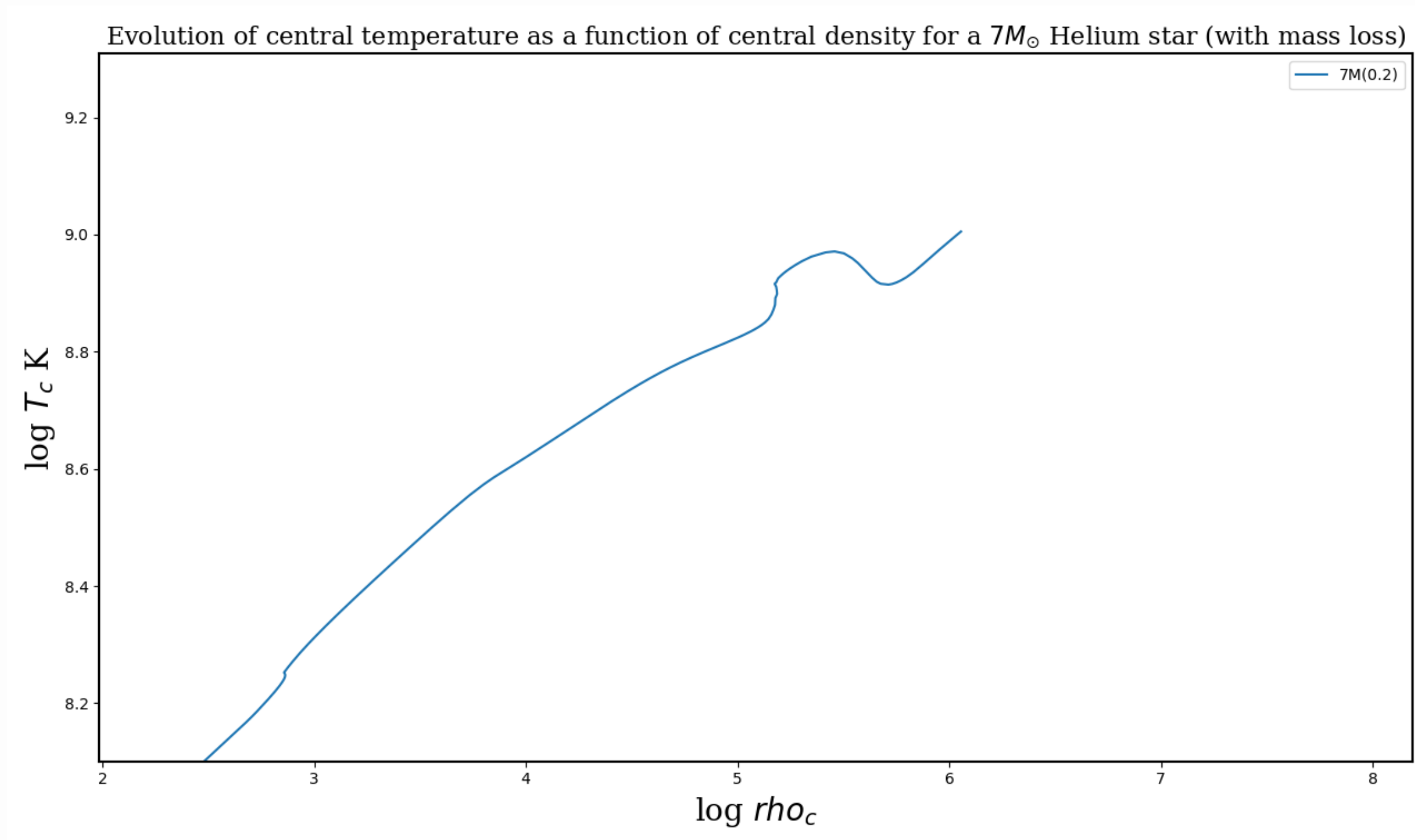


The evolutionary tracks in the central density–central temperature plane of a 3-, 5- and 7- $M_{\odot}$ helium star

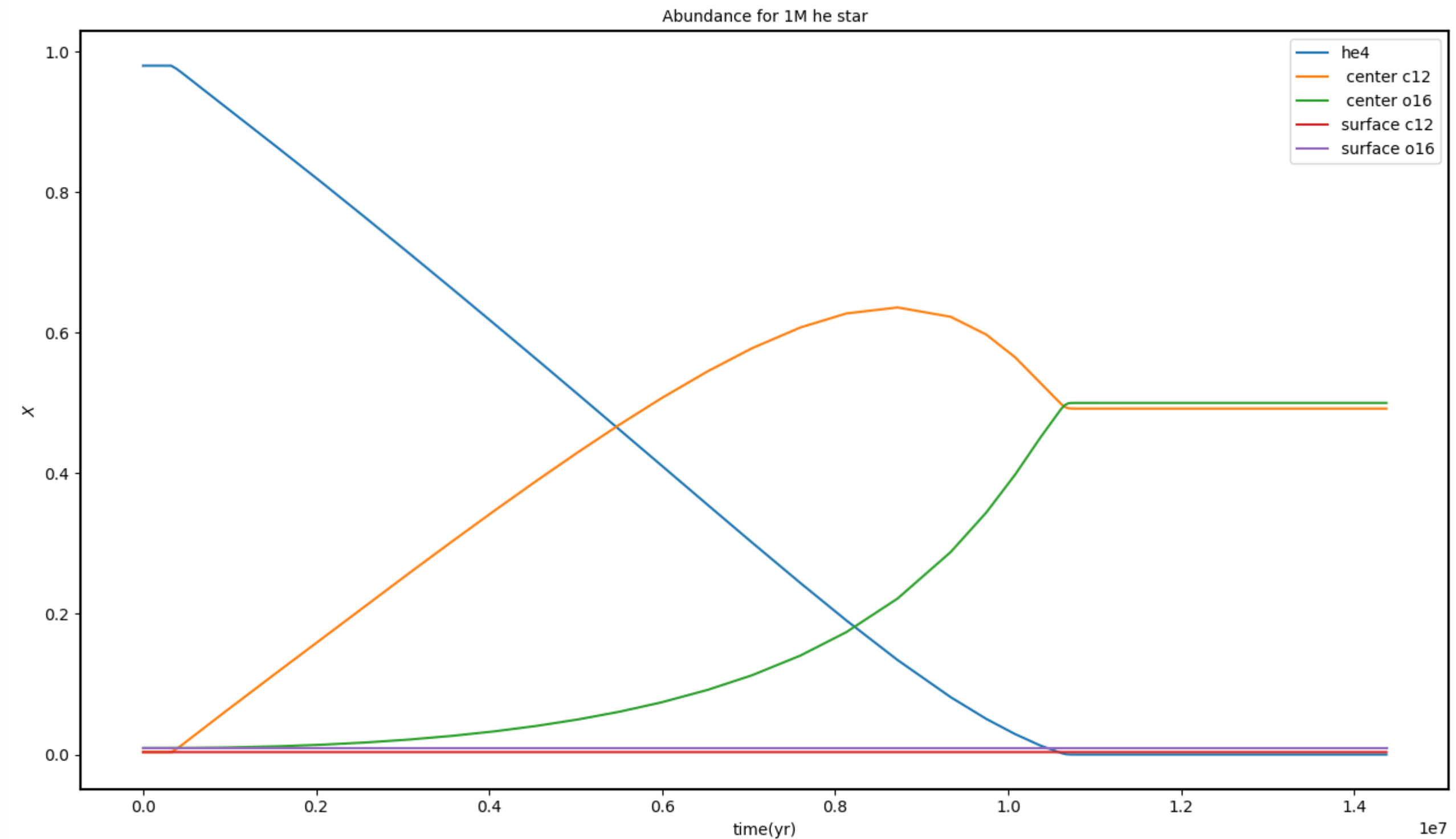


The letters marked along the evolutionary track correspond to the following:

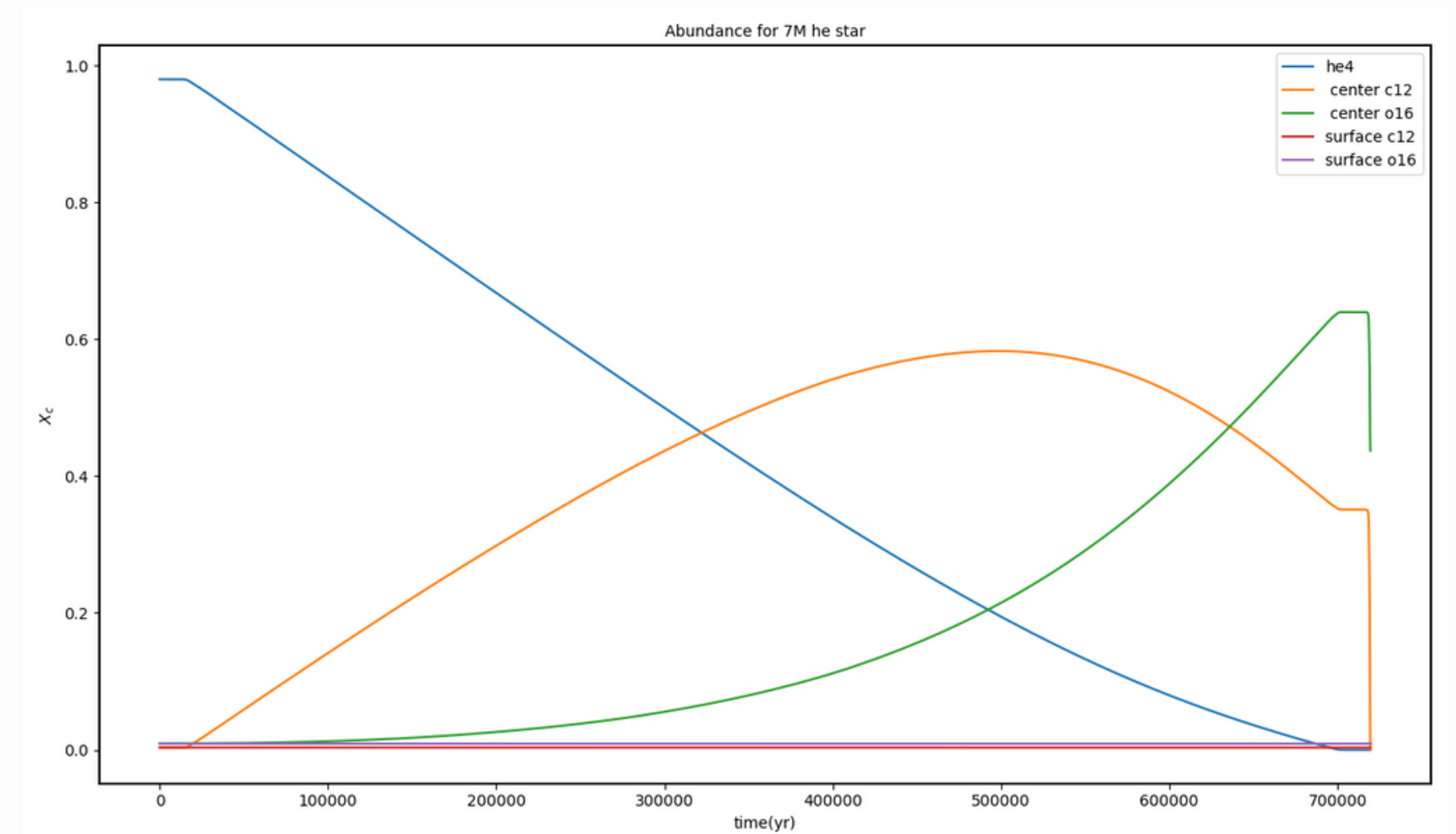
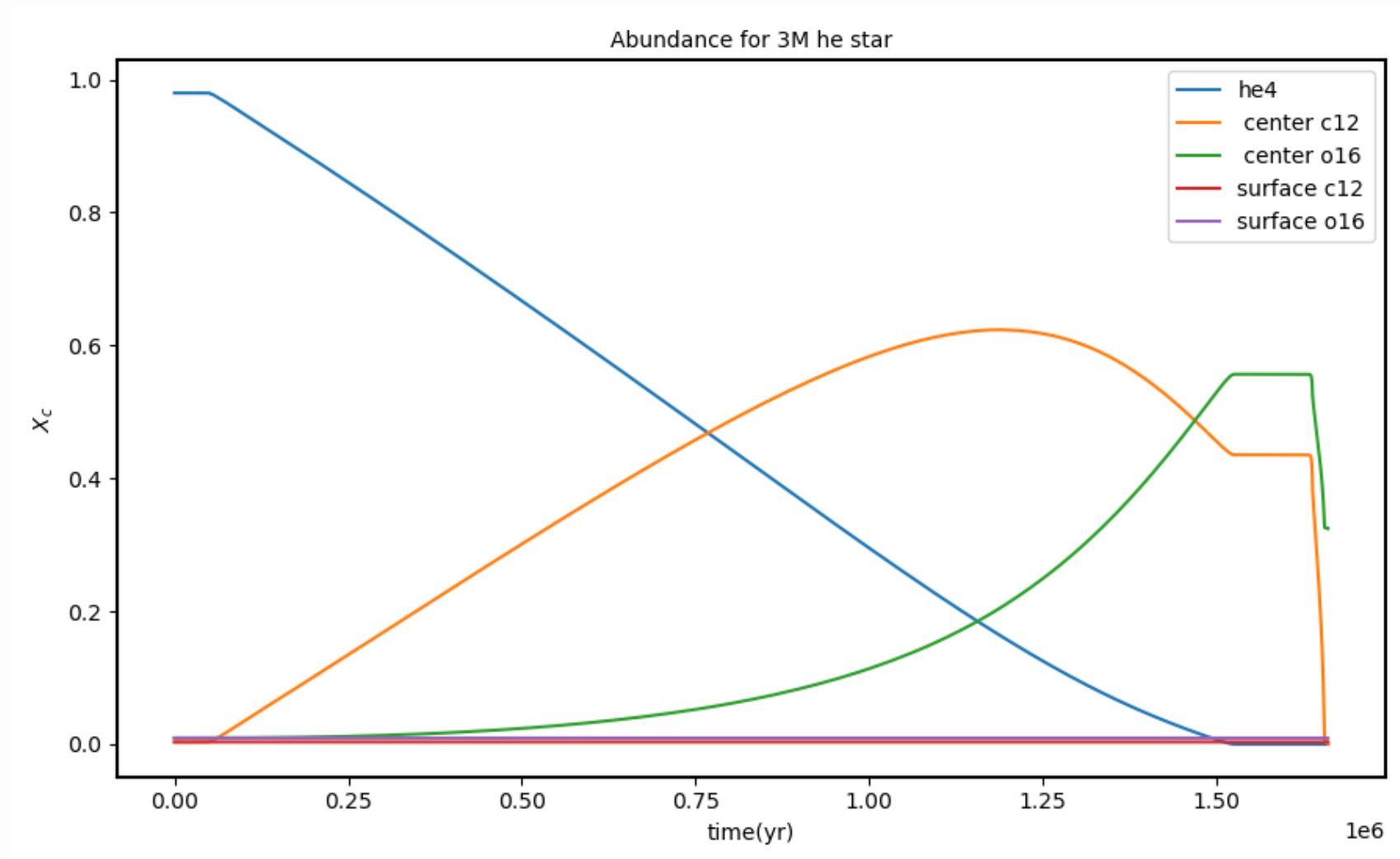
a- carbon core burning, b(& c)- exhaustion (& depletion) of carbon in the core, d- start of convective carbon shell burning, e- end of convective carbon shell burning



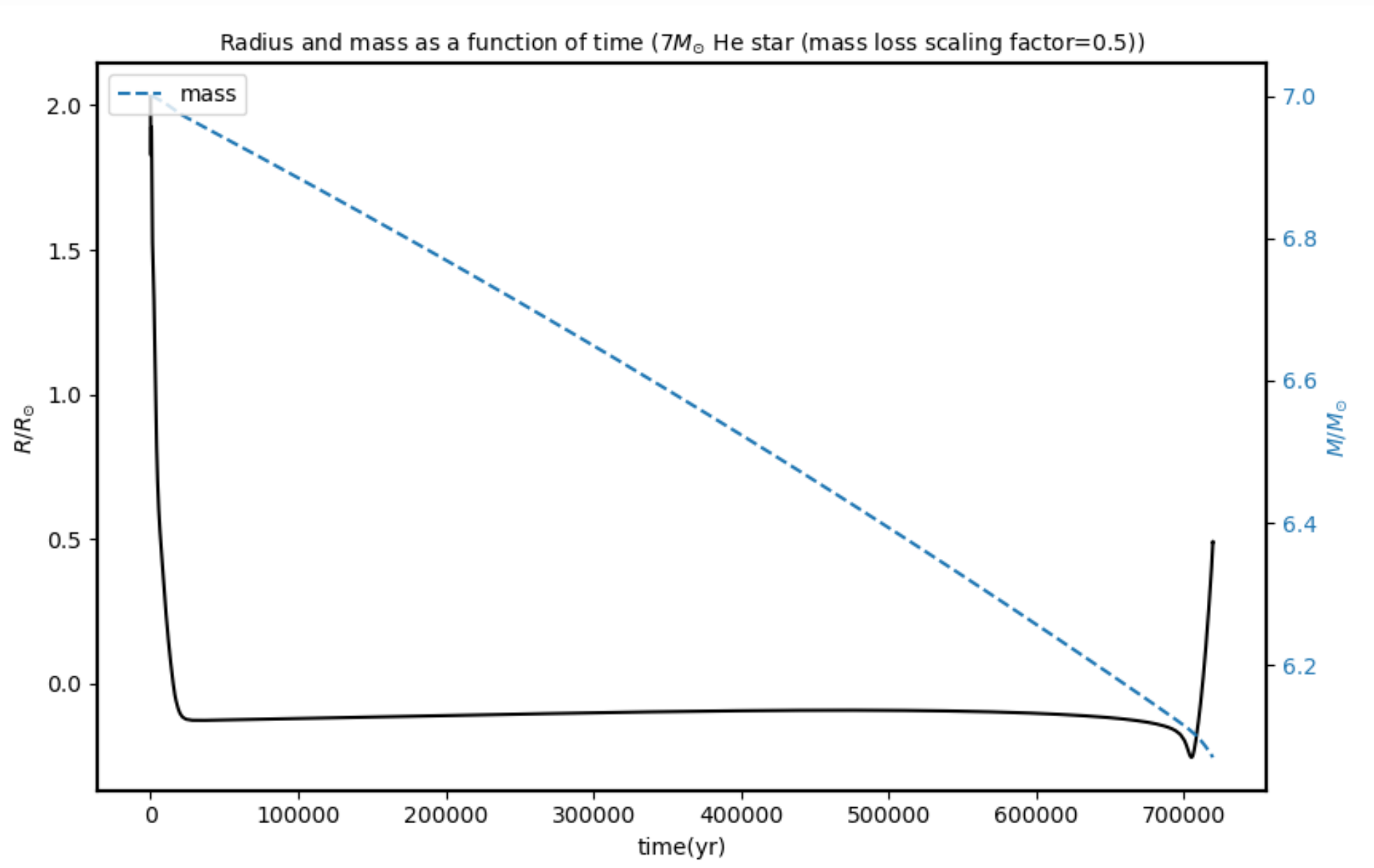
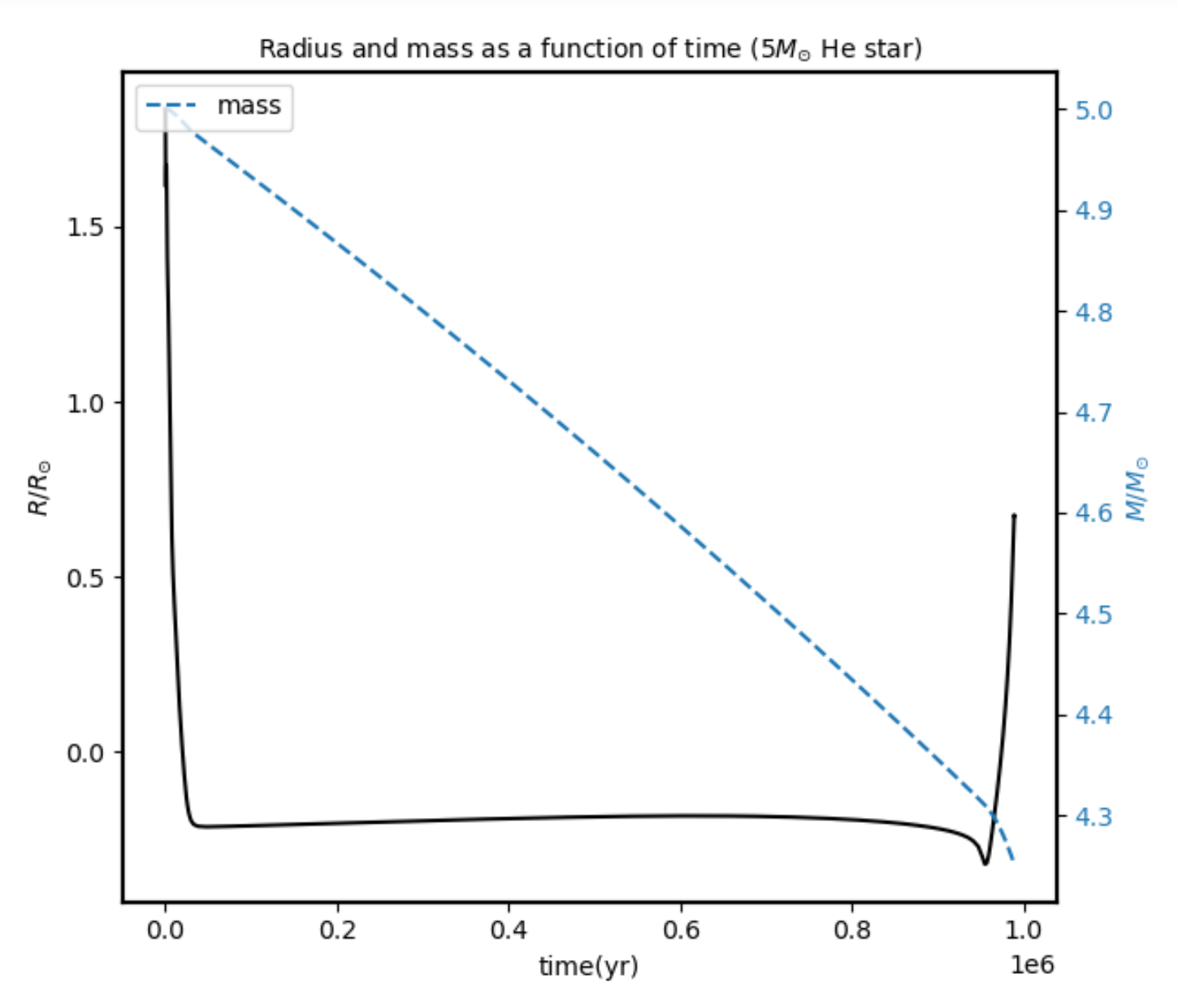
He, C, O abundances of a solar mass helium star



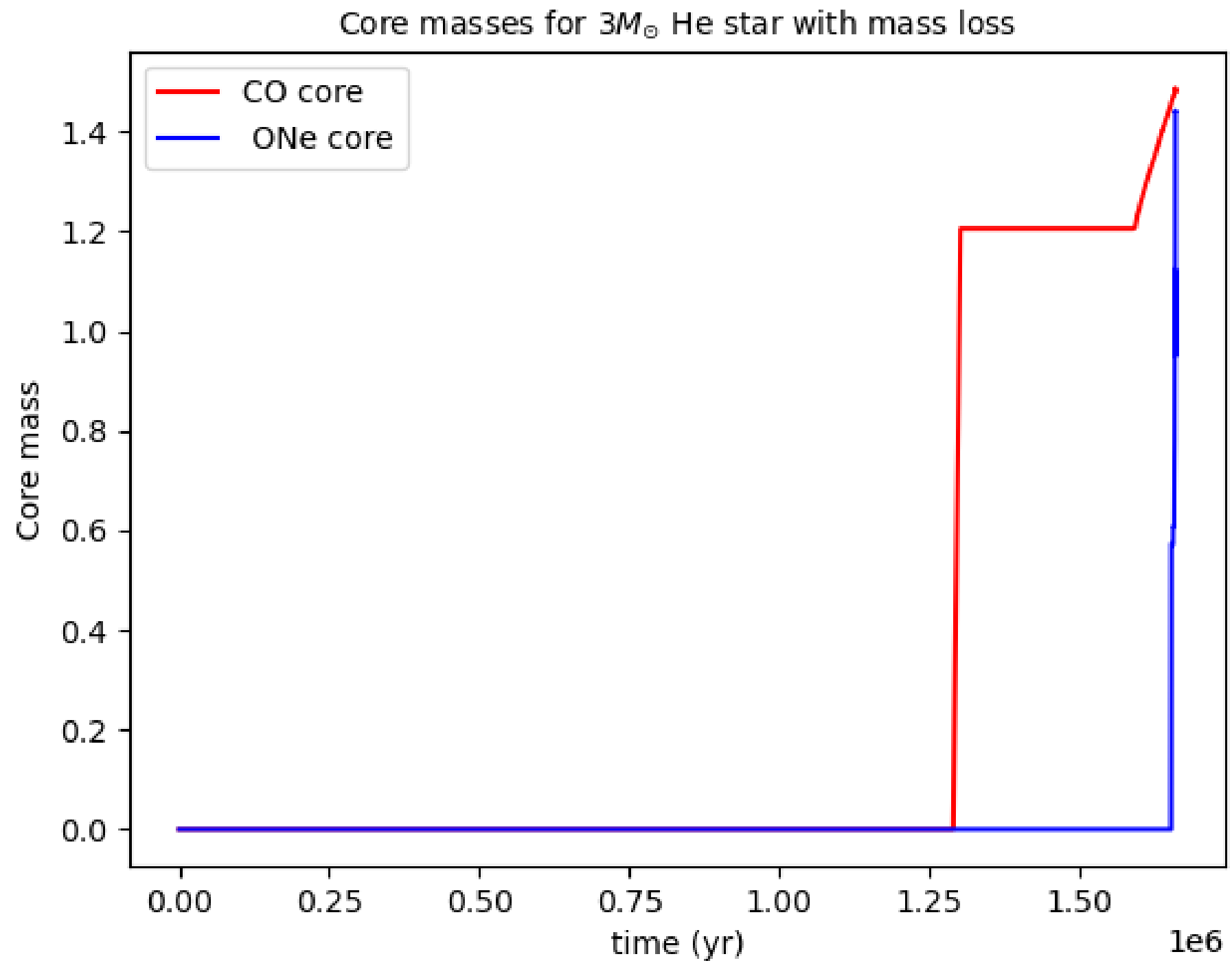
He, C, O abundances of 3- and 7 solar mass helium star

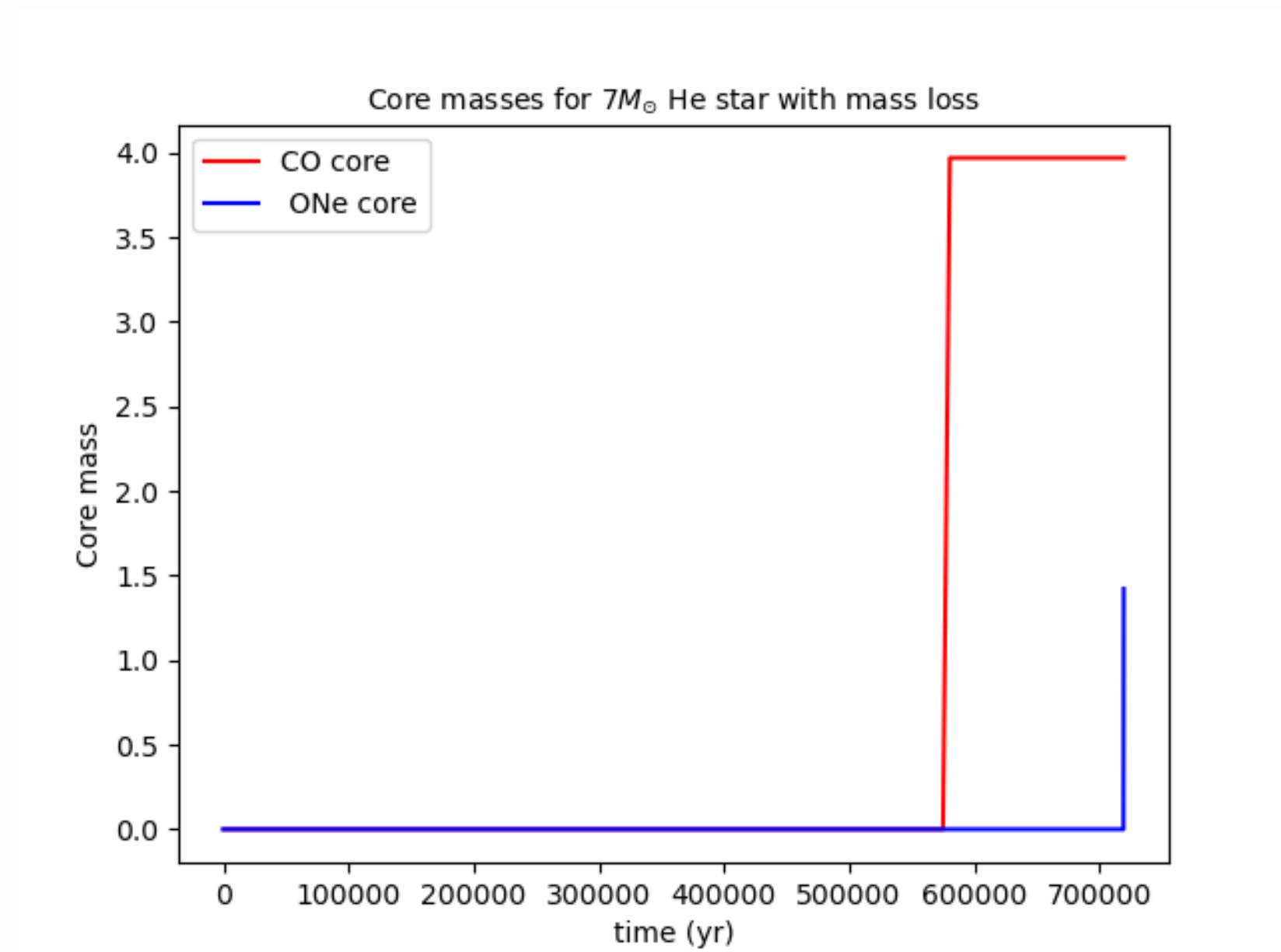
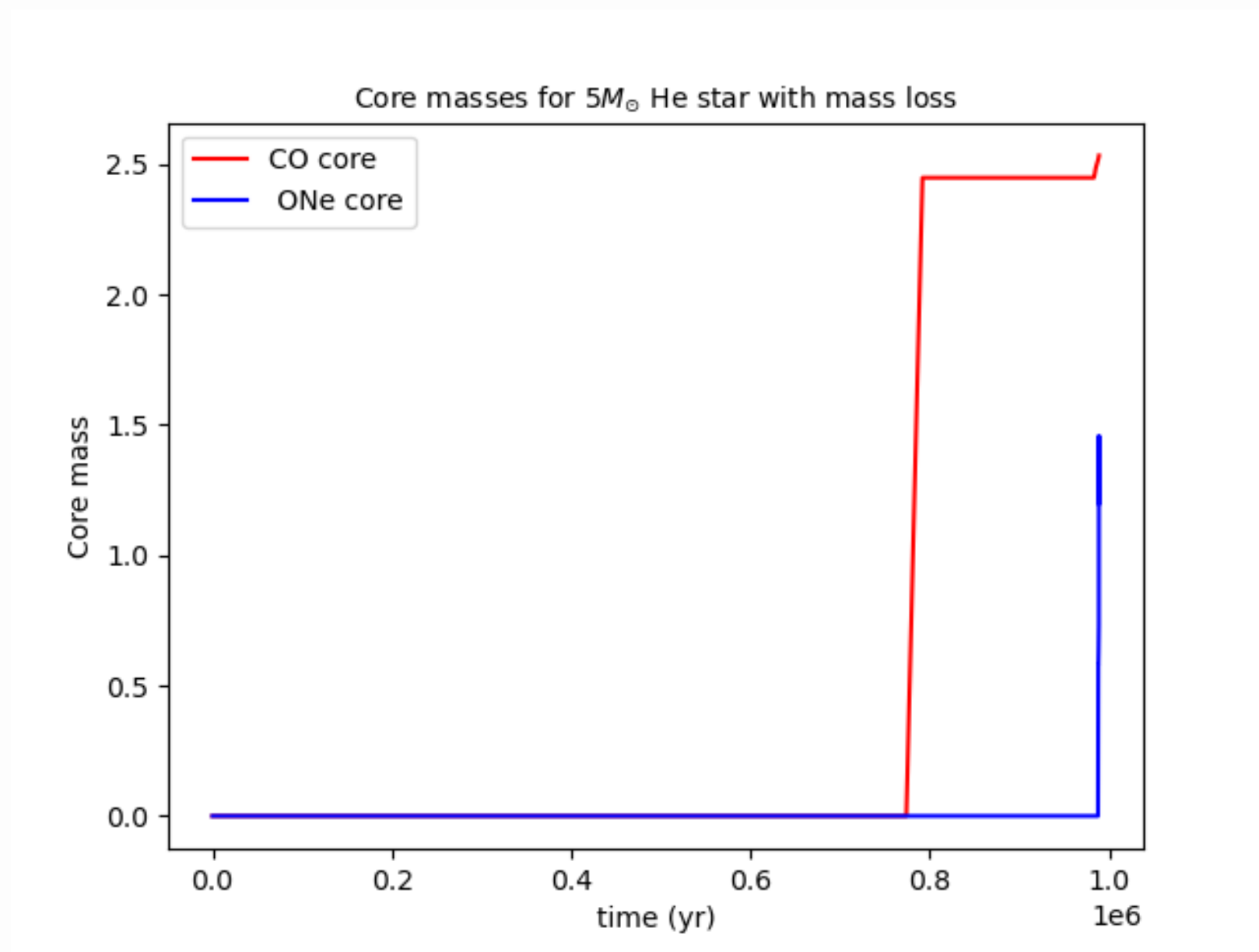


Variation of Radius and Mass of a 5- and 7 solar mass helium star with time



CO and ONe core masses





fixed !

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